

Stochastic Processes

The Two Key Components of a Stochastic Process

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- 1 **The State Space:** What are the possible values the random variable can take at any given time?
- 2 **The Time Parameter:** How is the process observed over time? Is it at discrete steps or continuously?

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- **Example:** The temperature at a specific location over a 24-hour period.

Combining Time and State Space

We can create a classification by combining the type of time parameter and the type of state space. Let's look at some examples for continuous time:

- **Continuous Time, Discrete State Space**

- Let $X(t)$ be the number of incoming calls to a switchboard in the interval $[0, t)$.
- Time t is continuous, but the state space (the count of calls) is discrete: $\{0, 1, 2, \dots\}$.

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- **Continuous Time, Continuous State Space**

- Let $X(t)$ be the maximum temperature at a location in the interval $[0, t)$.
- Time t is continuous, and the state space (the temperature) is also continuous.

The Four General Types of Processes

Based on the nature of the time parameter and the state space, one-dimensional stochastic processes can be classified into four general types:

- (i) Discrete time, discrete state space**
 - *Example: Result of a die roll at each throw.*
- (ii) Discrete time, continuous state space**
 - *Example: Temperature measured at noon every day.*
- (iii) Continuous time, discrete state space**
 - *Example: Number of customers in a store at time t .*
- (iv) Continuous time, continuous state space**
 - *Example: The price of a stock at time t .*

Standardizing the Notation

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The parameter is usually an integer, n .

- Notation:**

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Author's Convention

For simplicity, this text will use the notation $\{X(t), t \in T\}$ for **both** discrete and continuous cases. The nature of the parameter set T will be specified whenever necessary.

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Alternative Terminology

Some authors make a distinction based on the time parameter:

- A discrete parameter family is called a **stochastic sequence**.
- A continuous parameter family is called a **stochastic process**.

(Though often, "stochastic process" is used to refer to both.)

Relationship Between Variables in a Process

The Most Important Question

For a family of random variables $\{X_n, n \geq 1\}$, are the members $X_1, X_2, X_3, \dots$ independent of each other?

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 - For example, the temperature at 3 PM is highly dependent on the temperature at 2 PM.
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Key Idea

The relationship—or the dependency structure—among the random variables in the family is often of great importance and is a primary focus of study.

Summary

- We can use a unified notation $\{X(t), t \in T\}$ to describe any stochastic process.
- The parameter t is an index that often represents time, but can be other quantities.
- The most crucial feature of a stochastic process is typically the **relationship** among its constituent random variables over time.
- Unlike simple collections of random variables, in a stochastic process, this **dependency** is the interesting part!